COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S2
DATE PERFORMED:	August 25, 2026	**DATE SUBMITTED:**	August 25, 2026
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ACTIVITY NO. 4.1 **Ansible Basics**

1. PROCEDURE OUTPUT

```
jbbpios@workstation:~/ansible-lab$ ansible all -m apt -a update_cache=true --bec
ome --ask-become-pass
BECOME password:
```

```
jbbpios@workstation:~/ansible-lab$ ansible all -m apt -a name=vim-nox --become -
-ask-become-pass
BECOME password:
```

```
}
jbbpios@workstation:~/ansible-lab$ ansible all -i info.yaml -m apt -a name=vim-nox --become --ask-beco
me-pass
BECOME password:
server1 | FAILED! => {
    "msg": "Incorrect sudo password"
}
jbbpios@workstation:~/ansible-lab$ ansible all -i info.yaml -m apt -a name=vim-nox --become --ask-beco
me-pass
BECOME password:
[WARNING]: Platform linux on host server1 is using the discovered Python interpreter at
/usr/bin/python3.10, but future installation of another Python interpreter could change the meaning
of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
server1 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "cache_update_time": 1787670941,
    "cache_updated": false,
    "changed": true,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nTh
e following packages were automatically installed and are no longer required:\n  libreoffice-ogltrans
linux-headers-6.8.0-40-generic\n  linux-hwe-6.8-headers-6.8.0-40 linux-hwe-6.8-tools-6.8.0-40\n  linux
-image-6.8.0-40-generic linux-modules-6.8.0-40-generic\n  linux-modules-extra-6.8.0-40-generic linux-t
ools-6.8.0-40-generic\nUse 'sudo apt autoremove' to remove them.\nThe following additional packages wi
ll be installed:\n  fonts-lato liblua5.2-0 libruby3.0 rake ruby ruby-net-telnet ruby-rubygems\n  ruby-
webrick ruby-xmlrpc ruby3.0 rubygems-integration vim-common vim-runtime\n  vim-tiny\nSuggested package
s:\n  ri ruby-dev bundler cscope vim-doc indent\nThe following NEW packages will be installed:\n  font
s-lato liblua5.2-0 libruby3.0 rake ruby ruby-net-telnet ruby-rubygems\n  ruby-webrick ruby-xmlrpc ruby
3.0 rubygems-integration vim-nox vim-runtime\nThe following packages will be upgraded:\n  vim-common v
im-tiny\n2 upgraded, 13 newly installed, 0 to remove and 289 not upgraded.\nNeed to get 14.7 MB/17.9 M
B of archives.\nAfter this operation, 75.7 MB of additional disk space will be used.\nGet:1 http://ph.
archive.ubuntu.com/ubuntu jammy-updates/main amd64 vim-tiny amd64 2:8.2.3995-1ubuntu2.35 [707 kB]\nGet
:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 vim-common all 2:8.2.3995-1ubuntu2.35
[81.5 kB]\nGet:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-xmlrpc all 0.3.2-1u
buntu0.1 [24.9 kB]\nGet:4 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libruby3.0 amd6
4 3.0.2-7ubuntu2.13 [5116 kB]\nGet:5 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 vim-
```

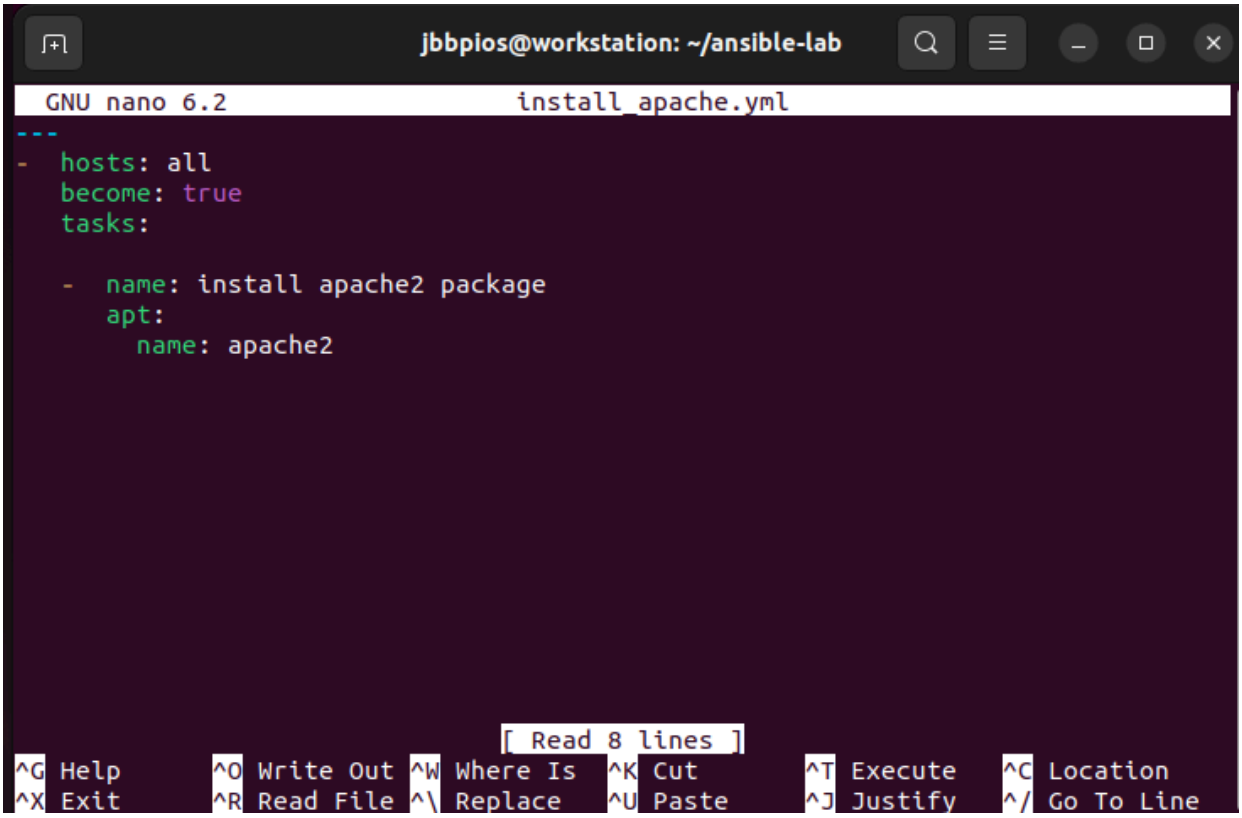
```
jbbpios@workstation:~/ansible-lab$ ansible all -i info.yaml -m apt -a name=vim-nox --become --ask-beco
me-pass
BECOME password:
[WARNING]: Platform linux on host server1 is using the discovered Python interpreter at
/usr/bin/python3.10, but future installation of another Python interpreter could change the meaning
of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "cache_update_time": 1787670941,
    "cache_updated": false,
    "changed": false
}
```

```
jbbpios@workstation:~/ansible-lab$ cd /var/log
jbbpios@workstation:/var/log$ ls
alternatives.log    bootstrap.log  dmesg.4.gz       kern.log          ubuntu-advantage.log
alternatives.log.1  btmp           dpkg.log         kern.log.1        ufw.log
apt                 btmp.1         dpkg.log.1       kern.log.2.gz     ufw.log.1
auth.log            cups           faillog          lastlog           ufw.log.2.gz
auth.log.1          dist-upgrade   fontconfig.log   openvpn           unattended-upgrades
auth.log.2.gz       dmesg          gdm3             private           wtmp
boot.log            dmesg.0        gpu-manager.log  speech-dispatcher
boot.log.1          dmesg.1.gz     hp               syslog
boot.log.2          dmesg.2.gz     installer        syslog.1
boot.log.3          dmesg.3.gz     journal          syslog.2.gz
jbbpios@workstation:/var/log$ cd apt
jbbpios@workstation:/var/log/apt$ cat history.log

Start-Date: 2026-08-13  08:23:58
Commandline: apt install python3-pip -y
Requested-By: jbbpios (1000)
Install: libalgorithm-merge-perl:amd64 (0.08-3, automatic), manpages-dev:amd64 (5.10-1ubuntu1, automat
ic), g++-11:amd64 (11.4.0-1ubuntu1~22.04.3, automatic), gcc-11:amd64 (11.4.0-1ubuntu1~22.04.3, automat
ic), build-essential:amd64 (12.9ubuntu3, automatic), libctf-nobfd0:amd64 (2.38-4ubuntu2.12, automatic)
, libtsan0:amd64 (11.4.0-1ubuntu1~22.04.3, automatic), libpython3-dev:amd64 (3.10.6-1~22.04.1, automat
ic), g++:amd64 (4:11.2.0-1ubuntu1, automatic), gcc:amd64 (4:11.2.0-1ubuntu1, automatic), libalgorithm-
diff-perl:amd64 (1.201-1, automatic), libbinutils:amd64 (2.38-4ubuntu2.12, automatic), libfakeroot:amd
64 (1.28-1ubuntu1, automatic), binutils-x86-64-linux-gnu:amd64 (2.38-4ubuntu2.12, automatic), libcc1-0
:amd64 (12.3.0-1ubuntu1~22.04.3, automatic), zlib1g-dev:amd64 (1:1.2.11.dfsg-2ubuntu9.2, automatic), p
ython3-wheel:amd64 (0.37.1-2ubuntu0.22.04.1, automatic), libfile-fcntllock-perl:amd64 (0.22-3build7, a
utomatic), dpkg-dev:amd64 (1.21.1ubuntu2.6, automatic), libasan6:amd64 (11.4.0-1ubuntu1~22.04.3, autom
atic), libnsl-dev:amd64 (1.3.0-2build2, automatic), rpcsvc-proto:amd64 (1.4.2-0ubuntu6, automatic), ma
ke:amd64 (4.3-4.1build1, automatic), libctf0:amd64 (2.38-4ubuntu2.12, automatic), lto-disabled-list:am
d64 (24, automatic), python3-dev:amd64 (3.10.6-1~22.04.1, automatic), libcrypt-dev:amd64 (1:4.4.27-1,
automatic), libpython3.10-dev:amd64 (3.10.12-1~22.04.16, automatic), binutils-common:amd64 (2.38-4ubun
tu2.12, automatic), libitm1:amd64 (12.3.0-1ubuntu1~22.04.3, automatic), python3-pip:amd64 (22.0.2+dfsg
-1ubuntu0.7), libdpkg-perl:amd64 (1.21.1ubuntu2.6, automatic), python3.10-dev:amd64 (3.10.12-1~22.04.1
6, automatic), libc-dev-bin:amd64 (2.35-0ubuntu3.14, automatic), libc-devtools:amd64 (2.35-0ubuntu3.14
, automatic), python3-distutils:amd64 (3.10.8-1~22.04, automatic), libc6-dev:amd64 (2.35-0ubuntu3.14,
automatic), libjs-jquery:amd64 (3.6.0+dfsg+~3.5.13-1, automatic), libquadmath0:amd64 (12.3.0-1ubuntu1~
```

```
jbbpios@workstation:~/ansible-lab$ ansible all -i info.yaml -m apt -a name=snapd
 --become --ask-become-pass
BECOME password:
[WARNING]: Platform linux on host server1 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "cache_update_time": 1787670941,
    "cache_updated": false,
    "changed": false
}
```

```
jbbpios@workstation:~/ansible-lab$ ansible all -i info.yaml -m apt -a "name=snap
d state=latest" --become --ask-become-pass
BECOME password:
[WARNING]: Platform linux on host server1 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
server1 | SUCCESS => {
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    },
    "cache_update_time": 1787670941,
    "cache_updated": false,
    "changed": false
}
```




```
jbbpios@workstation:~/ansible-lab$ nano install_apache.yml
jbbpios@workstation:~/ansible-lab$ ansible-playbook -i info.yaml --ask-become-pa
ss install_apache.yml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Platform linux on host server1 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

TASK [install apache2 package] *************************************************
ok: [server1]

PLAY RECAP *********************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

```
ss update_cache.yml
BECOME password:

PLAY [all] ********************************************************************

TASK [Gathering Facts] ********************************************************
[WARNING]: Platform linux on host server1 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

TASK [update repository index] ************************************************
changed: [server1]

TASK [install apache2 package] ************************************************
ok: [server1]

PLAY RECAP ********************************************************************
server1                    : ok=3    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

```
LAY [all] *********************************************************************

ASK [Gathering Facts] *********************************************************
[WARNING]: Platform linux on host server1 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
ttps://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
k: [server1]

ASK [update repository index] *************************************************
hanged: [server1]

ASK [install apache2 package] *************************************************
k: [server1]

ASK [add PHP support for apache] **********************************************
k: [server1]

LAY RECAP *********************************************************************
erver1                     : ok=4    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. What is the importance of using a playbook?

Playbooks are important because they allow system administration tasks to be automated and executed in a structured and repeatable way. Instead of running multiple ad hoc commands manually, a playbook defines the desired configuration and tasks that Ansible will perform on the managed nodes.

2. Summarize what we have done in this activity.

In this activity, we used elevated Ansible ad hoc commands to update package caches and install packages such as vim-nox and snapd on remote servers. We then created and executed an Ansible playbook to automate the installation of Apache, update the package cache, and add additional package support.

Conclusions/Learnings

Through this activity, we learned how Ansible can remotely manage servers using elevated privileges through --become and --ask-become-pass. We also learned the importance of inventory files in identifying managed nodes and YAML playbooks in automating multiple system administration tasks. Overall, the activity demonstrated how Ansible makes server management more efficient, consistent, and easier to automate.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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